

INTERNSHIP REPORT

ON

“INTRODUCTION TO DATA SCIENCE”

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

Submitted By

ADITYA PANDAY

Roll Number: 2410031676

Batch: 2026–27

CANDIDATE’S DECLARATION

I, ADITYA PANDAY, hereby declare that the internship report titled “Introduction to Data Science” has been prepared by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. The report presents the learning and understanding gained during the internship/training. I have acknowledged the sources used in preparing this report and confirm that this report has not been submitted to any other institute for any degree or diploma requirement.

Signature: _____

Student Name: ADITYA PANDAY

Roll No.: 2410031676

ACKNOWLEDGEMENT

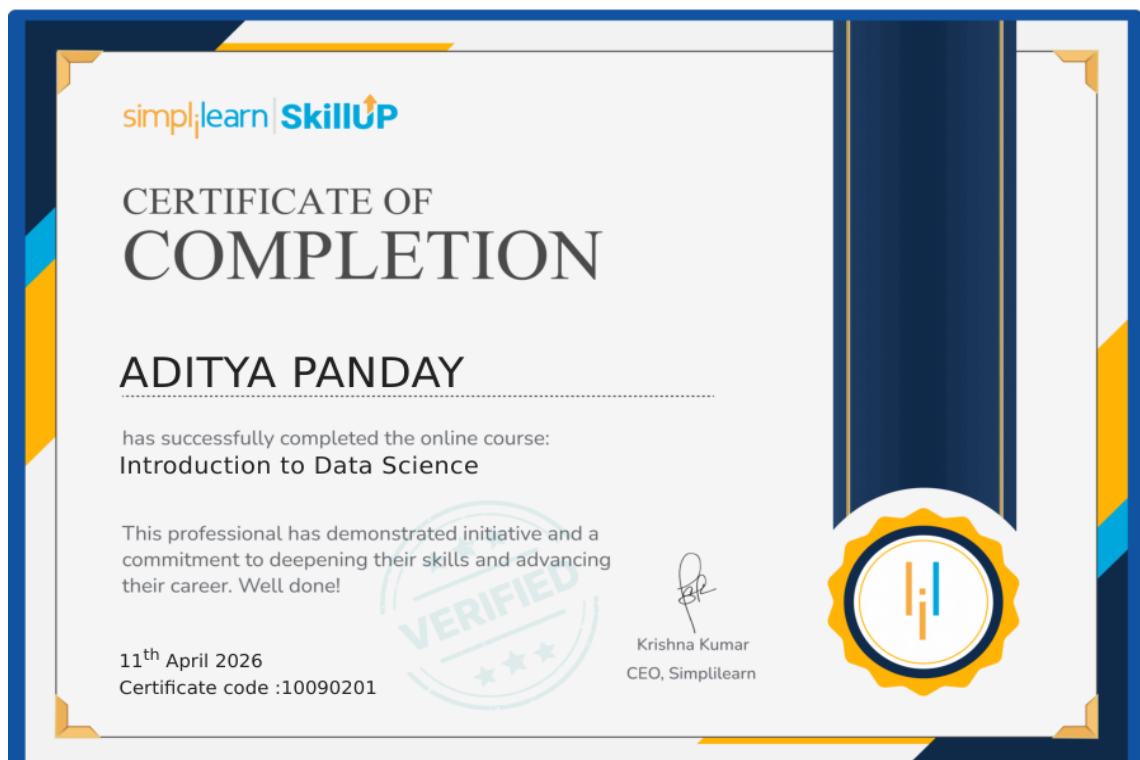
I express my sincere gratitude to everyone who supported me during my internship/training on “Introduction to Data Science”. This learning experience helped me develop a structured understanding of the data science field and its practical workflow. I am thankful to the faculty and mentors whose guidance and encouragement helped me complete this report. I also express my gratitude to my parents for their continuous support and blessings.

Signature: _____

Student Name: ADITYA PANDAY

Roll No.: 2410031676

INTERNSHIP COMPLETION CERTIFICATE



4. PROJECT DESCRIPTION

4.1 Introduction

Data Science is an interdisciplinary field focused on working with data to discover useful information, patterns and insights. The “Introduction to Data Science” internship/training provided an introductory foundation for understanding how data can be collected, prepared, explored and interpreted for meaningful analysis. The learning area connects concepts from statistics, programming, data handling and analytical thinking.

4.2 Organization / Training Profile

The certificate supplied for this report identifies the learning programme as “Introduction to Data Science”. The certificate is issued in the name of ADITYA PANDAY and is dated 11th April 2026. The certificate code is 10090201. The supplied certificate does not provide additional organization or provider details, so no unsupported organizational information has been added to this report.

4.3 Problem Statement

Modern applications generate large amounts of data, but raw data by itself does not directly provide useful conclusions. The central problem addressed by an introductory data science workflow is how to transform raw or unorganized data into a form that can be explored, analyzed and communicated clearly. This requires understanding data, identifying useful patterns and presenting findings in a meaningful way.

4.4 Project Objectives

- Understand the basic concept and role of Data Science.
- Understand the general data science workflow from data to insights.
- Develop familiarity with data preparation and exploratory analysis concepts.
- Understand how analytical results can support decision-making.
- Build a foundation for further learning in data analytics, machine learning and related areas.

4.5 Scope of the Project

The scope of this introductory work covers fundamental Data Science concepts and the general analytical lifecycle. It provides a foundation for understanding data collection, data preparation, exploratory analysis, visualization and interpretation. The supplied certificate only establishes the internship/training title and completion date; therefore, specific advanced modules or organization-specific tools are not claimed unless supported by additional documentation.

4.6 Technologies and Tools Used

The supplied certificate does not specify particular software tools, programming languages or platforms. Accordingly, this report does not attribute any specific tool to the internship as a verified requirement. The conceptual areas relevant to an Introduction to Data Science programme include data handling, basic statistical thinking, exploratory analysis and data visualization.

4.7 System Architecture

A conceptual Data Science workflow can be represented as: Data Sources → Data Collection → Data Cleaning & Preparation → Exploratory Data Analysis → Analysis/Modeling → Visualization & Interpretation → Insights/Decision Support. This architecture describes the general flow of a data science task rather than a claim about a specific system implemented during the certificate programme.

4.8 Methodology

The methodology followed for understanding a data science problem can be organized into sequential stages. First, the problem and required data are identified. Next, data is collected and checked for quality. The data is then cleaned and prepared for analysis. Exploratory analysis is performed to understand distributions, relationships and patterns. Appropriate analytical methods can then be applied, followed by visualization and interpretation of the results. Finally, the findings are communicated in a clear form.

4.9 Expected Outcomes

The expected learning outcomes of an introductory Data Science programme include a clear understanding of the Data Science lifecycle, improved ability to reason about data, awareness of data preparation and exploratory analysis, and a foundation for progressing toward data analytics and machine learning. The internship also supports the development of analytical and problem-solving skills.

4.10 Certificate and Communication Proof

The internship completion certificate is attached as supporting proof. It records the name ADITYA PANDAY, the programme title “Introduction to Data Science”, the date 11th April 2026, and certificate code 10090201.

5. BIBLIOGRAPHY / REFERENCES

[1] IILM University, School of Computer Science and Engineering, Internship Report Format 2026–27.

[2] Internship completion certificate supplied by the student: “Introduction to Data Science”, dated 11th April 2026, Certificate Code 10090201.